Quantum Information Processing (2025) 24:37
https://doi.org/10.1007/s11128-025-04655-3

Enhancing quantum approximate optimization with CNN-CVaR integration

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Received: 22 May 2024 / Accepted: 8 January 2025 / Published online: 24 January 2025
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Abstract

The quantum approximate optimization algorithm (QAOA) represents a promising approach for tackling combinatorial optimization challenges on near-term quantum devices. Central to QAOA optimization is the minimization of the expectation of the problem Hamiltonian for parameterized trial quantum states, which motivates the exploration of advanced optimization techniques. In this study, we propose a novel combinatorial optimization strategy, CNN-CVaR-QAOA, which integrates a convolutional neural network (CNN) with conditional value at risk (CVaR) to optimize QAOA circuits. By replacing the traditional loss function with CVaR and leveraging CNN for variational quantum parameter optimization, we demonstrate the superior efficacy of CNN-CVaR-QAOA through experimental validation on Erdos–Renyi random graphs. Our results show better solutions across various graph configurations. Furthermore, we investigate the influence of the CVaR parameter (α) on algorithm performance, revealing that lower α values lead to smoother objective functions and improved approximation ratios. This work indicates that CNN-CVaR-QAOA offers significant advantages in optimizing QAOA parameters, particularly in the context of near-term intermediate-scale quantum era, highlighting its potential to enhance QAOA optimization efforts across diverse optimization domains.

Keywords Convolutional neural network · Conditional value at risk · QAOA optimization

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1 Introduction

With the advent of near-term intermediate-scale quantum (NISQ) computing, the landscape of computational problem-solving is undergoing a profound transformation. Quantum computers are emerging as powerful tools poised to surpass classical counterparts in tackling complex real-world problems. Combinatorial optimization challenges, including Max-Cut, Graph Coloring, Traveling Salesman, and Scheduling Management, find their solutions through Ising Hamiltonians, thus making them ripe for quantum computational approaches [1–4].

Hybrid quantum–classical algorithms, particularly the quantum approximate optimization algorithm (QAOA), stand out as promising methodologies in this arena. By leveraging the principles of variation, QAOA offers an avenue to approximate solutions for problems encoded by Hamiltonians [5]. Recent advancements in both experimental implementations and theoretical underpinnings have propelled QAOA into the limelight [4, 6–14]. The tantalizing prospect of quantum advantage over classical algorithms beckons, with evidence mounting to support its universality and computational prowess [8, 15–18].

Despite these advancements, the QAOA still faces several limitations and potential issues hindering its effectiveness. Notably, the trade-off between circuit depth and performance, constrained by coherence times in existing and near-term quantum processors, poses a significant challenge [19–22]. Additionally, the complexity introduced by the QAOA ansatz, with its increasing number of variational parameters, presents challenges for classical optimizers [23–25]. Previous studies have shown that the optimal QAOA parameters exhibit specific patterns [26, 27], leading to depth-sequential strategies and machine learning-based methods for parameter initialization [10, 23, 28–30]. For instance, Alam et al. adopted a regression model to predict high-depth parameters from low-depth ones [26], while Amosy et al. applied iterative neural networks for parameter initialization, selecting the most promising parameter sets from cluster centers for the given problem [31]. However, existing machine learning approaches exhibit limitations in universality, necessitating numerous models for different problem sizes or QAOA depths, resulting in higher costs for preparing extensive training data for deeper QAOAs [26, 28]. As described in reference [28], convolutional neural networks (CNNs) have been employed to predict the parameters of the QAOA. Xie et al. utilized a residual neural network to predict the QAOA parameters and demonstrated its efficiency and advantages compared to existing classical optimizers. The advantage of this approach lies in the ability to find the optimal solution of the QAOA more quickly by using CNNs to predict the parameters. Additionally, it can automatically learn parameter settings that adapt to different problem instances during the training process. This provides a new potential method for solving quantum optimization problems. Furthermore, relying on the mean of expectation values as the objective function can hinder the optimization process [6, 32, 33]. Therefore, it is crucial to explore effective objective functions that dynamically control the expressiveness and trainability of the QAOA for a given combinatorial problem. As described in reference [6], Barkoutsos et al. proposed that a new CVaR function is proposed as a replacement for the objective function, which greatly improves the quality of our

optimized quantum circuits by dynamically smoothing the objective function through the variation of parameter.

In this study, we propose a novel method, CNN-CVaR-QAOA, which synergistically integrates a classical convolutional neural network (CNN) with conditional value at risk (CVaR) aggregation functions to effectively optimize variational parameters, thereby augmenting the performance of the QAOA. The adoption of the CNN architecture stems from the intrinsic similarities between pixel properties in images and QAOA parameters, such as their continuity and correlation among adjacent values. CNNs can be understood as learning the prior graph structure, training on a certain number of graph example problems, and then applying the trained network for parameter prediction, swiftly predicting the best parameters for the next depth of a given graph example problem, maintaining good performance even when dealing with larger problem instances. In comparison with RNN optimization, the latter consists of an iterative quantum–classical optimization loop between classical processing units (CPU) and quantum processing units (QPU). From an optimization perspective, the RNN is tasked with finding a set of parameters to minimize this cost function, often finding an approximate minimum. Given the stochastic nature of the function output, acting as a noiseless unbiased estimate of the true output values poses a significant challenge for optimizing the variational quantum algorithm, as each iterative optimization requires calling the quantum circuit for computation, significantly increasing the computational burden [30]. Regarding graph neural networks (GNNs) structures [34], GNNs are a type of neural network model designed specifically for handling graph-structured data, typically operating through a message-passing process within the graph. GNNs operate by encoding the input graph and outputting transformed encodings, with each graph node initially encoded as a vector that is then updated by the GNNs to include relevant feature information about the nodes in the graph. However, the scalability of GNNs may be limited for larger problem scales [35]. By employing CVaR instead of minimizing the expectation value, our model iteratively optimizes by minimizing the CVaR, thereby aligning more closely with the objective of solving combinatorial optimization problems. Comparative evaluations conducted on Max-Cut problem instances on Erdos–Renyi random graphs of different sizes demonstrate that our proposed CNN-CVaR-QAOA method outperforms alternative approaches, particularly at lower depths. This innovative framework not only advances the frontier of quantum optimization but also underscores the potential of synergistic integration between classical and quantum techniques in addressing complex computational challenges.

The paper is structured as follows: Sect. 2 provides an introduction to the quantum approximate optimization algorithm. In Sect. 3, we outline the implementation details of the CNN-CVaR method. Section 4 elaborates on the numerical and experimental study, followed by a conclusion where potential future steps are discussed.

2 Quantum approximate optimization algorithm

The quantum approximate optimization algorithm (QAOA) stands as a pivotal solution tailored for tackling combinatorial optimization (CO) dilemmas, leveraging a blend of quantum and classical methodologies. Illustrated in Fig. 1, QAOA harnesses a param-

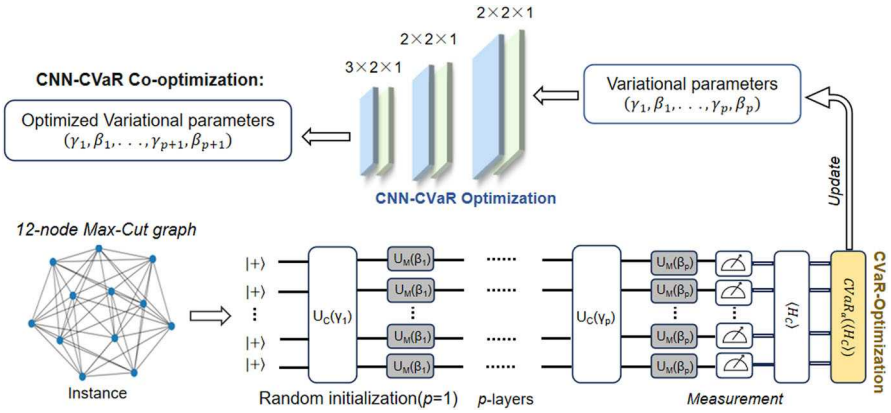

Fig. 1 A schematic diagram illustrating a p-level quantum approximate optimization algorithm along with the utilization of CNN-CVaR optimization. An example showcasing the Max-Cut problem on a 12-node graph, with a quantum circuit taking input. Alternate application of $U_C(\beta)$ and $U_M(\gamma)$, followed by the measurement of the final state to obtain the expectation value relative to the objective function H_C. We utilize the QN-SPSA optimizer to randomly initialize the generation of the starting variational parameters ($p = 1$). Subsequently, CVaR optimization is employed to replace the objective function, and the resulting output is fed into a convolutional neural network to determine the optimal parameters γ and β

eterized quantum circuit denoted as $U(\theta)$, colloquially referred to as a variational form, to produce trial wavefunctions. The architecture of a p-layer circuit crafts states through the expression:

$$|\phi(\gamma, \beta)\rangle = \left(\prod_{p=1}^{p} e^{-i\beta_p H_M} e^{-i\gamma_p H_C} \right) |+\rangle^{\otimes N} , \tag{1}$$

where $|+\rangle$ represents the eigenstate, the problem Hamiltonian H_C is used to encode the total energy of the system, and H_M is known as the mixer Hamiltonian, given by $H_M = \sum_{n=1}^{N} \delta_n^X$, where δ_n^X is the Pauli-X operator acting on the qubit n [36]. Finally, the p-dimensional vectors β and γ serve as the variational parameters corresponding to H_M and H_C, respectively.

The QAOA operates by measuring the parameterized quantum state to derive a solution, with its parameter adjustment process guided by a classical optimization algorithm. The primary objective is to identify the optimal variational parameters (β^*, γ^*) that minimize the objective function $F(z)$:

$$F(z) = \langle \phi(\gamma, \beta) | H_C | \phi(\gamma, \beta) \rangle . \tag{2}$$

This study predominantly addresses the Max-Cut problem. In formal terms, let $G(V, E)$ represent an undirected graph with a vertex set $V = N$, and E representing the edge set. The problem attempts to partition the nodes of a graph G into two sets $[+1, -1]$, maximizing the number of edges between nodes from these two sets. The

optimal solution z^* for Max-Cut maximizes the objective function $C(z)$:

$$C(z) = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(1 - (-1)^{z_i}(-1)^{z_j}), \tag{3}$$

where z represents a bit string. Theoretical studies have proven that determining z^* is an NP-hard problem, necessitating approximation techniques in most scenarios. A commonly used solver for Max-Cut is the Goemans–Williamson (GW) algorithm, which employs semidefinite programming (SDP) methodologies to achieve an approximation ratio of 0.879 [37].

To tackle combinatorial problems using physical systems, it is imperative to map the problem onto a Hamiltonian, where the ground state corresponds to the optimal solution [38]. In the case of the Max-Cut problem with n nodes, the cost Hamiltonian H_C is constructed using the Pauli-Z operators as follows:

$$H_C = \frac{1}{2} \sum_{(i,j)\in E} w_{ij}(I - \delta_i^Z \delta_j^Z). \tag{4}$$

Here, we simplify by setting all edge weights to 1, allowing us to confine each γ and β within the intervals $[0, \pi]$ and $[0, \pi/2]$, respectively. For the Max-Cut problem, the commonly used performance metric for QAOA is the approximation ratio R, which is defined as:

$$R = \frac{F(z)}{C_{\max}}, \tag{5}$$

where $C_{\max}$ is the maximum cut of the graph.

3 CNN-CVaR optimization

In this section, we propose to utilize CNN as a classical network parameter prediction architecture to achieve more efficient optimization of QAOA [39], with CVaR as the objective function [40]. The architecture of the proposed CNN model and the aggregation function for CVaR optimization are elucidated below.

In this experiment, we utilized the noiseless state vector simulator of the Qiskit platform to execute all QAOA circuits [41]. In this simulated environment, we defined the number of measurements for the quantum circuit to be 1024 to ensure data adequacy and the reliability of statistical results. We have picked the problem graphs for Max-Cut-QAOA from the Erdos–Renyi ensemble [42] using Python NetworkX package [43]. Our CNN model aims to learn a mapping function from the parameters of p-depth QAOA to the parameters of $p+1$-depth. The p-depth and $p+1$-depth parameters are denoted as tensors of size $1 \times 2 \times p$ and $1 \times 2 \times p + 1$, respectively. We initialize the input by randomly setting the parameters of a QAOA circuit with a depth of 1 using the QN-SPSA optimizer, a stochastic optimizer belonging to the family of gradient descent methods [44]. As shown in Fig. 1, our CNN model architecture comprises two key components: up-sampling and down-sampling. The up-sampling segment of our

CNN model is responsible for extracting features from the input QAOA parameters. This segment consists of two convolutional layers, both employing a stride of 1 and ReLU activation function [45] as the activation function. Each layer utilizes a 2×2 kernel and provides one zero-padding for all four inputs. The first layer expands the input dimensions to 16, which are further augmented to 64 dimensions by the second layer. On the other hand, the down-sampling section involves simplifying the extracted features. Here, a single convolutional layer with a $64 \times 3 \times 2$ filter is employed to reduce the dimensions from 64 to 1. This results in the final output of the CNN network, which produce the final optimized variational parameters for the QAOA circuit. During the training of the CNN, we pretrained all embeddings in the dataset to initialize the network parameters. For training, we set the batch size to 6 and conducted 50 epochs of training iterations. To optimize the network parameters, we employed the Adam optimizer with a learning rate set to 0.0001. Throughout the training process, we utilized mean squared error (MSE) as the loss function to measure the disparity between predicted values and actual values. We continuously adjusted the network weights to minimize this error, thereby enhancing the predictive performance of the network.

The objective of CVaR is to prioritize improving the best outcomes observed, rather than solely relying on the sample expectation value as the traditional optimization objective function. First, we prepare the trial wave function $|\psi(\gamma, \beta)\rangle$ on a quantum processor. Subsequently, we measure the quantum bits to obtain an n-bit string $x_0 \ldots x_n$. Each observed string can be readily converted into a sample of $\langle\psi(\gamma, \beta)|H|\psi(\gamma, \beta)\rangle$ since H is a weighted sum of the tensor products of the Bubbly z-matrix, and each such term can be computed with a simple parity check. The method of CVaR that we adopt involves selecting a set of observation results within measurements, where the sample H_k is sorted in nondecreasing order, taking $[H_1, \ldots, H_k]$ as the objective function [33]. Specifically, assuming H_k is the energy corresponding to a computational basis vector, and ordering them such that a larger k corresponds to a larger energy. For each state, if a state repeats the measurement K times, there are (at most) K different values of H_k. Simultaneously, the CVaR parameter $\alpha \in (0, 1]$ is introduced, where α represents further calculating the expectation value within the α proportion of $[H_1, \ldots, H_k]$, thereby smoothing the objective function. The CVaR function, denoted as CVaR_α, is defined as follows:

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha K} \sum_{k=0}^{\alpha K} H_K, \tag{6}$$

where $\alpha = 1$ corresponds to the sample mean expectation value. Decreasing α biases the selection of observation samples increasingly toward the best outcomes. It is crucial to note that employing CVaR optimization modifies our objective function (2) to:

$$\min(\mathrm{CVaR}_\alpha(F(z))), \tag{7}$$

CVaR_α considers only a subset of measurements, leading to an estimator variance of $\mathrm{O}(1/(\mathrm{K}\alpha^2))$, where K is the number of samples. As α decreases, CVaR increasingly

emphasizes the best observed samples while yielding a smoother and more manageable objective function. This enhanced smoothness facilitates more efficient convergence in optimization algorithms.

However, as indicated in Ref. [33], lowering the value of α to increase the approximate ratio of QAOA results in performance degradation compared to the case of variational quantum eigensolver (VQE). This improvement may stem from the fact that the outcomes derived from using CVaR instead of the original objective function tend to be more focused on achieving better results as α decreases, as described by Eq. (6), thereby rendering the objective function relatively smooth [6]. Simultaneously, employing the original random initialization prediction method may hinder us from achieving superior parameter prediction results within a few iterations. In such scenarios, our CNN-CVaR co-optimization is anticipated to address this issue.

4 Performance of CNN-CVaR optimization on QAOA

4.1 Comparison between our heuristics and other optimizations

The analysis above indicates that the utilization of the CNN architecture for optimizing subsequent layer parameters within the QAOA was observed. These parameters are incrementally predicted based on initial values. The CVaR aggregation function is employed to optimize the cost function for parameter updates, thereby enhancing the efficacy of optimal measurements through dynamic cost function adjustments. By amalgamating these methodologies with QAOA, we effectively address combinatorial optimization challenges, resulting in the development of our proposed approach, termed CNN-CVaR-QAOA. The validation of our approach is conducted through the resolution of the Max-Cut problem on regular graphs. We commence this validation process by delineating our experimental setup, encompassing the configuration of hyperparameters and the construction of the Max-Cut graph. Subsequently, to assess the proficiency of our proposed initialization methodology, we juxtapose the performance of our optimization strategy against standard strategies such as random initialization (RI) and interpolation (INTERP) [10].

To thoroughly evaluate the performance of our proposed CNN-CVaR-QAOA, we assembled a diverse collection of graphs with varying sizes, node degrees, and edge probabilities. For analysis, we employed Erdos–Renyi (ER) graphs, randomly allocating both degree and edge probability for each node. Multiple instances were generated for each configuration to derive statistical outcomes using the Max-Cut solver, with the average approximation ratio as the primary metric. In our experimental setup, for QAOA with a depth of 1, we consistently used the QN-SPSA optimizer to compute initial parameters conducive to convergence.

Firstly, we employed CNN-CVaR-QAOA to analyze various graph configurations. In this experiment, we set the number of measurements for the quantum circuit to 1024. According to the principles of the CVaR equation, the parameter α has a significant impact on the experimental results. Specifically, when α is small, the CVaR equation focuses more on the better observed results, biasing the final outcome toward these better observations. If we choose a very small value for α, the approximation ratio

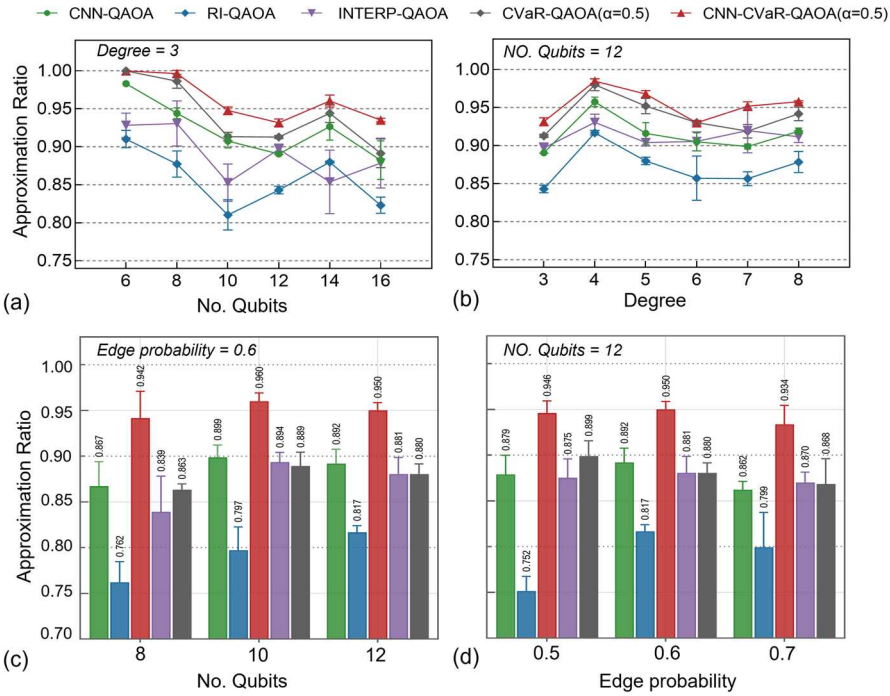

Fig. 2 The average performance of QAOA is depicted on the graph. **a** On 3-regular ER graphs, the approximation ratio obtained by each optimization method is presented as a function of the number of nodes(qubits) in the graph, with the number of layers p set to 3. **b** On 12-node ER graphs, the approximation ratio obtained by each optimization method is represented as a function of the number of degrees in the graph, with the layer count **p** set to 3. **c** On ER sampled graphs with edge probability of 0.6, comparison of the relative performance of CNN-CVaR method with four other optimization methods on graphs contains 8, 10, and 12 qubits. **d** Comparison of the relative performance of the CNN-CVaR approach to four other optimization methods on ER sampled graphs with edge probabilities is set to 0.5, 0.6, 0.7 for 12 qubits

of the experimental results will tend toward 1, indicating that we are almost exclusively focusing on the optimal observed results, which may lead to biased evaluations. Conversely, if α is set too large, it may include too many unfavorable observations, resulting in our method showing less significant improvement in approximation ratio when compared to other state-of-the-art methods, thereby failing to demonstrate the advantages of our approach. Therefore, considering the accuracy of the experimental results and the comparative effectiveness with other state-of-the-art methods, we set the value of α to 0.5. This setting effectively captures better observed results while appropriately including some unfavorable observations, enabling our method to demonstrate a more significant improvement in approximation ratio when compared to other state-of-the-art methods, thereby proving the effectiveness and competitiveness of our approach.

The numerical outcomes are visualized in Fig. 2, presenting the results derived from CNN-CVaR-QAOA ($\alpha = 0.5$) alongside four heuristic Max-Cut solvers, namely

convolutional neural network prediction (CNN-QAOA), random initialization (RI-QAOA), heuristic Interpolation method (INTERP-QAOA), and conditional value at risk optimization (CVaR-QAOA). Each scenario encompasses three distinct graph instances, where the number of nodes fluctuates within the range of [6,16], mirroring the number of quantum bits in the quantum circuit. The degree of nodes varies within the interval [3,8], and the probability of edge formation fluctuates within the interval [0.5,0.7]. To gauge the performance gap between CNN-CVaR-QAOA ($\alpha = 0.5$) and the optimal classical outcomes, we used the approximate ratio of each result as a measure, where the method that includes CVaR replaces the original cost function with it and calculates the expectation value of the circuit in the next iteration of optimization. Subsequently, we use Eq. (5) to generate an approximate ratio. For circuit depth 3, we validated our approach across diverse regular graphs, as shown in Fig. 2a, b, affirming the superior approximation ratio achieved by our CNN-CVaR-QAOA ($\alpha = 0.5$) strategy compared to the other four algorithms as qubit count and node degrees increased. Especially, the CNN-CVaR co-optimization significantly improves performance over the CVaR optimization of QAOA. Throughout this process, we consistently utilize the final expectation values of the circuits under each optimization method as the criterion for assessing the discrepancy between them and the optimal cut value.

Additionally, with circuit depth set at 2 and α at 0.5, we evaluated the algorithm across different configurations of ER random edge probability graphs, as indicated in Fig. 2c, d. Our CNN-CVaR co-optimization method improved approximation ratio values in the range of $0.06-0.2$ compared to other four methods. The findings consistently underscored the optimal performance of CNN-CVaR-QAOA ($\alpha = 0.5$) across all cases. Thus, our heuristic algorithm represents a substantial enhancement in variational parameter optimization given the available resources.

Then, we conducted a series of experiments utilizing benchmark instances from the Max-Cut dataset to investigate our optimization architecture by varying circuit depths. Through meticulous analysis, we aimed to elucidate performance nuances across different strategies and depths. Figure 3a depicts randomly generated 8-node 3-regular graphs, serving as the focal point for comparative assessments among five distinct methods. Notably, as illustrated in Fig. 3b, the CNN-CVaR-QAOA ($\alpha = 0.5$) strategy exhibited rapid convergence to the optimal solution at a depth of 3. The CVaR ($\alpha = 0.5$) method, without CNN, converges to the optimal result at the depth of 4. Further insights were obtained from Fig. 3c, d, revealing that CNN-CVaR-QAOA ($\alpha = 0.1$) achieved optimality at a depth of 4 for the same 12-node 3-regular graphs. Due to the setting of the α parameter, CVaR-QAOA ($\alpha = 0.1$) also converges to the optimal result at this depth, but in comparison with CNN-CVaR-QAOA ($\alpha = 0.1$), it trails in the initial depths. Notably, superior performance was observed on unweighted graphs, potentially due to the inherent complexities involved in optimizing solutions for weighted counterparts. Furthermore, the circuit depth emerged as a critical factor influencing the effectiveness of our strategy. Particularly significant is our approach's ability to produce comparable outcomes to alternative methods, even at shallower depths. For instance, in the 12-node 3-regular graphs scenario, the performance of CNN-CVaR-QAOA ($\alpha = 0.1$) at a depth of 2 rivaled that of CNN-QAOA and INTERP-QAOA at depths as deep as 7. This finding demonstrates the potential of our approach

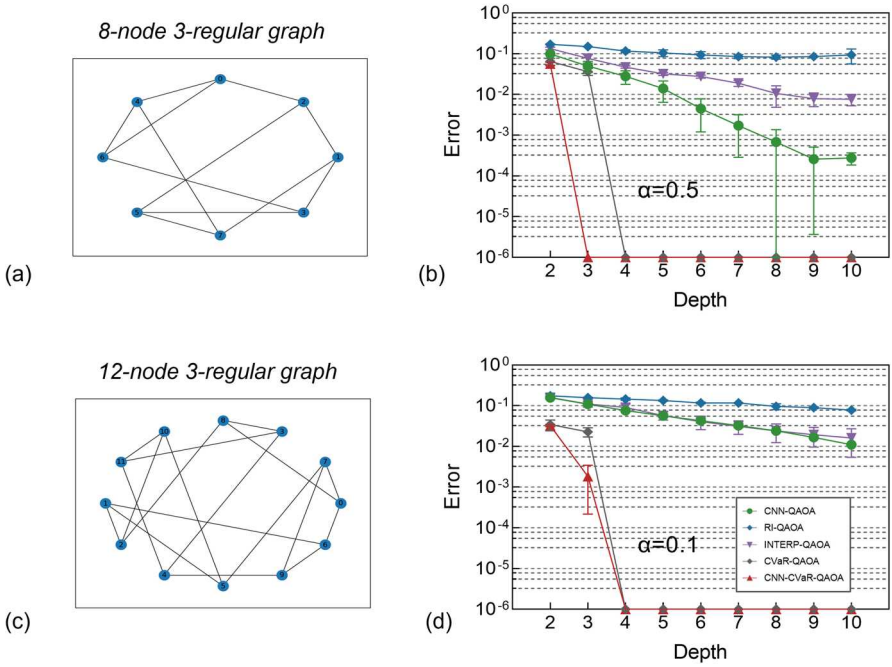

Fig. 3 **a** A set of diagrams extracted from the evaluation of an 8-node 3-regular graph instances. **b** On 8-node 3-regular graph instances, the errors during the optimization process of the CNN-CVaR ($\alpha = 0.5$) method are compared to four other QAOA optimization methods. The QAOA circuit varies from two layers to 10 layers. **c** A set of diagrams extracted from the evaluation of an 12-node 3-regular graph instances. **d** On 12-node 3-regular graph instances, the errors during the optimization process of the CNN-CVaR ($\alpha = 0.1$) method are compared to four other QAOA optimization methods. The QAOA circuit varies from two layers to 10 layers. The QAOA circuit varies from two layers to 10 layers

to efficiently leverage quantum resources while approximating optimal solutions for the Max-Cut problem.

In the updated experiments, we selected a quantum system corresponding to a 14-node, degree-3-regular graph for our study. Initially, as shown in Fig. 4a, we conducted tests at a circuit depth of 2. In this evaluation, we compared several approaches: random initialization, standalone CNN usage, interpolation method, combination of random initialization and CVaR, and the joint optimization strategy of CNN and CVaR, with the CVaR parameter alpha set to 0.5. The results demonstrated that optimizing the quantum system using the joint CNN-CVaR strategy, followed by running the quantum circuit with the optimized parameters, led to the lowest expectation value. A similar set of results is shown in Fig. 4b, which illustrates the expectation value differences achieved by various methods at a circuit depth of 3. Notably, the CNN-CVaR method converged to an expectation value very close to the theoretical minimum, further reinforcing the validity of our approach.

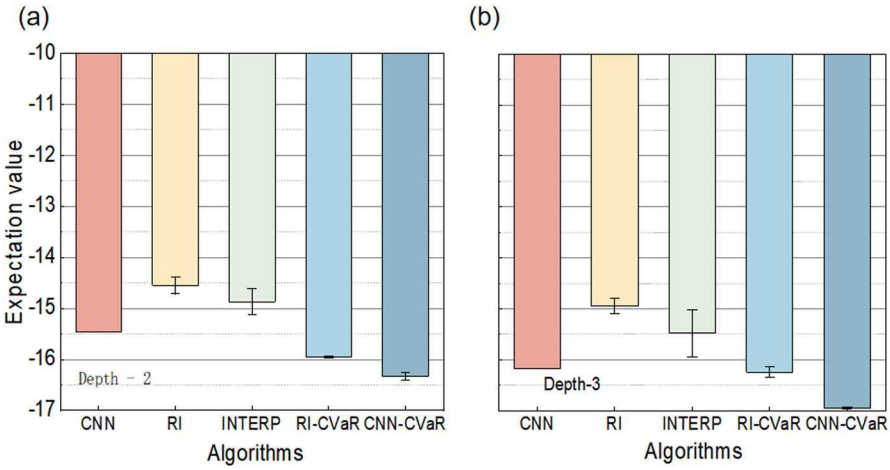

Fig. 4 The final converged ground state expectation values of the CNN-CVaR optimization method are compared with those of the other four methods on a 14-node, 3-regular graph. **a** Simulation on a quantum circuit of depth 2. **b** Simulation on a quantum circuit of depth 3

4.2 Analysis of hyperparameter in our co-optimization method

We examined the influence of hyperparameters, including the variational parameters γ and β, particularly the α parameter of a CVaR aggregation function, on the performance of the QAOA. We employed various configurations of regular graphs and random edge probability graphs while maintaining a quantum circuit with a fixed layer count of p=2. Across these setups, we obtain the results from three distinct non-isomorphic instances.

In Fig. 5, we present a heat map depicting the relationship between CVaR values and the parameters γ, β, both ranging from 0 to $\pi/2$, for various CVaR parameter α. As illustrated, the CVaR parameters are set to 1, 0.8, 0.5, and 0.1 on a 10-node 3-regular graphs, with $\alpha = 1$ representing to the unoptimized ansatz. By examining the convergence landscapes under various configuration scenarios, we can evaluate the quality of the results from the parameters. The yellow regions indicate the minimal values of the CVaR, signifying the convergence of the random 12-node 3-regular graph. It is evident that the positions of the minimum remain almost consistent as the parameter α changes, located in the ranges $\beta \in [0.2, 0.6]$ and $\gamma \in [0.3, 1.0]$, as indicated by the yellow dashed lines in Fig. 5. This demonstrates how the parameters β and γ distribute to achieve the optimal solutions for our CNN-CVaR co-optimization.

Regarding the CVaR parameter α, we systematically analyzed the impact of α parameter variations on the performance of the optimization system and examined whether this variation aligns with the logical evolution of the CVaR function. The CVaR values are computed using Eq. (6). Our research not only delved into the role of the α parameter in the quantum optimization process but also revealed how fine-tuning the sample inclusion proportion can optimize algorithm performance and enhance the quality of solutions. Through this perspective, we aim to further validate the effectiveness and reliability of our research findings. Our findings reveal that when a decrease in

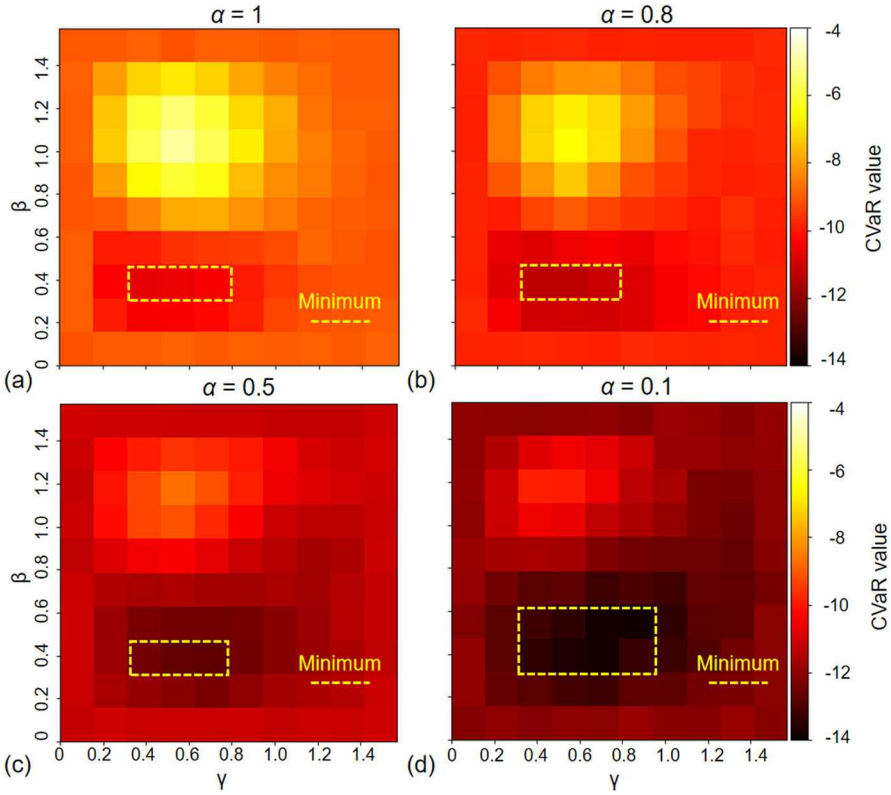

Fig. 5 The landscape of the CVaR value for a random 12-node 3-regular graph with $\gamma \in [0, \pi]$ and $\beta \in [0, \pi]$. **a** Convergence landscape of the parameters perturbations when $\alpha = 1$. **b** Convergence landscape of the parameters perturbations when $\alpha = 0.8$. **c** Convergence landscape of the parameters perturbations when $\alpha = 0.5$. **d** Convergence landscape of the parameters perturbations when $\alpha = 0.1$. As the value of α decreases, the minimum CVaR value achieved by our CNN-CVaR co-optimization is closer to the ground state energy. Conversely, higher α values may encompass less favorable measurement outcomes, leading to correspondences with higher CVaR values that could adversely impact the final optimization results, causing a lower probability of converging to the ground state

α to 0.1 is associated with significantly lower CVaR values compared to the other three scenarios. The optimal variational parameters distribution expands as α decreases, as shown in Fig. 5, $\beta \in [0.3, 0.5]$ and $\gamma \in [0.3, 0.65]$ for $\alpha = 1$ and 0.8, and $\beta \in [0.3, 0.6]$ and $\gamma \in [0.3, 1.0]$ for $\alpha = 0.5$ and 0.1. This phenomenon can be attributed to the fact that, with $\alpha = 0.1$, the minimum CVaR value achieved by our CNN-CVaR co-optimization is closer to the ground state energy. Conversely, higher α values correspond to higher CVaR values, indicating a lower probability of converging to the ground state. Additionally, as α decreases, the upper bound of the CVaR value also significantly decreases.

As illustrated in Fig. 6, in a quantum system with a circuit depth of 2, we used the expectation value of the entire circuit state as a metric and conducted detailed tests on an Erdos–Renyi graph with 14 nodes and a degree of 3. We employed our

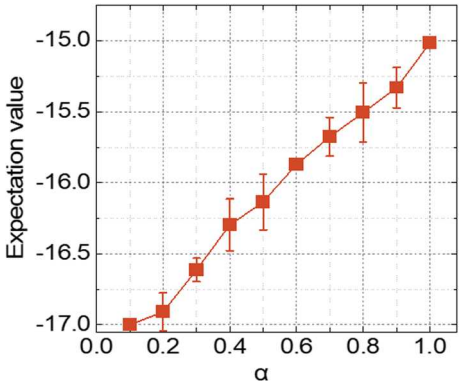

Fig. 6 In a configuration with a depth of 2, on a 14-node Erdos–Renyi graph with a degree of 3, as α varies from 0.1 to 1, the overall expectation values of the system circuits optimized for different α values using CVaR were observed. Each sampling result was obtained from more than three runs

strategy while varying α from 0.1 to 1, sampling the overall expectation values of the system corresponding to different α configurations. The experimental results clearly demonstrate that our proposed joint optimization strategy of CNN and CVaR yields promising outcomes on quantum systems with a circuit depth of 2. By measuring the final expectation value of the circuit state, we observed that as α decreases, the expectation values we sampled converge closer to the optimal expectation value. Moreover, saturation is achieved at $\alpha = 0.1$. The optimal expectation value mentioned in this text was computed using conventional brute-force enumeration methods [46]. These experimental findings robustly validate the efficacy of the method, providing solid experimental evidence for its reliability and practical prospects. These insights underscore the critical role of the CVaR parameter in the optimization process, particularly in the search for superior energy states.

Additionally, the quantitative performance analysis of CNN-CVaR co-optimization under different α parameters is shown in Fig. 7. The circuit depth p is set to 2, and on 10-node 3-regular graphs, we observed that under the parameter settings $\alpha = 0.1$, $\alpha = 0.3$, $\alpha = 0.5$, and $\alpha = 0.8$, the approximation ratio of CNN-CVaR-QAOA ($\alpha = 0.1$) compared to the parameters of $\alpha = 0.3$, $\alpha = 0.5$, $\alpha = 0.8$, and $\alpha = 1$ was on average improved by approximately 4.0%, 8.8%, 12.6%, and 13.7%, respectively. Similar improvements were observed on 12-node 3-regular graphs, 14-node 3-regular graphs, and 16-node 3-regular graphs. In Fig. 7b–d, more extensive testing was conducted on random regular graphs configurations 10-node 5-regular graphs, 12-node 5-regular graphs, 14-node 5-regular graphs, and 16-node 5-regular graphs, and on the random ER graphs, the edge probabilities are set to 0.5 and 0.7, while the number of nodes varies between 8 and 14, to assess the impact of α on the overall performance of CNN-CVaR-QAOA model. These observations further confirm the aforementioned conclusions, which indicate that when executing CNN-CVaR-QAOA on a larger-scale quantum system, a smaller value of α would lead to the superior performance. These improvements can be explained by Eq. (6), which shows that a smaller α value tends to favor optimal observation results, thereby smoothing the objective function.

Finally, to objectively evaluate the specific impact of the parameter α on the performance of the CVaR optimization system, we meticulously designed three different combinations of measurement states for quantum approximate optimization circuits

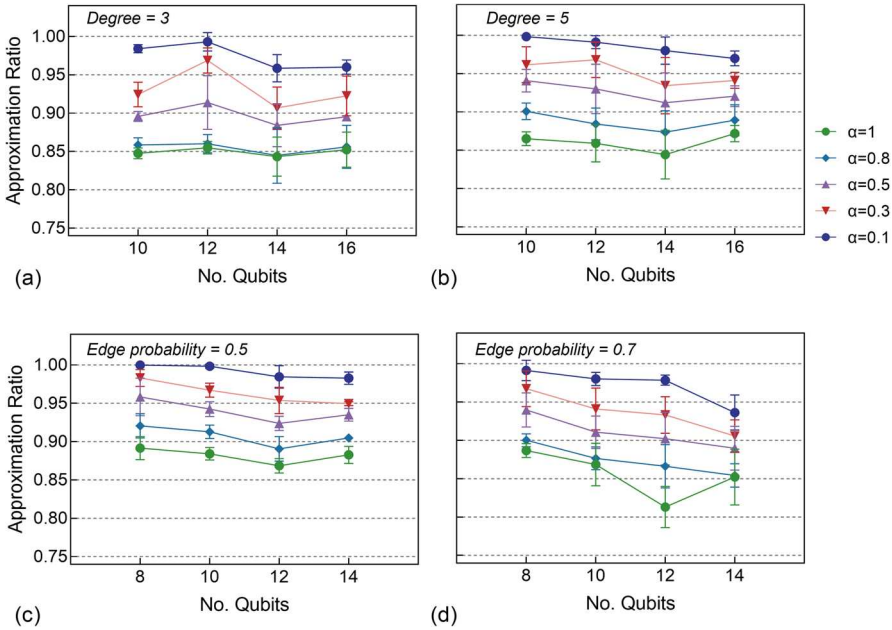

Fig. 7 The quantum approximation ratios obtained using classical simulation of CNN-CVaR-QAOA at different α values, with a depth of 2. **a** Comparing the impact of different α values on the overall performance for the Max-Cut problem on different random regular graphs with degree = 3. **b** Comparing the impact of different α values on the overall performance for the Max-Cut problem on different random regular graphs with degree = 5. **c** Assessing the effect of different α values on the overall performance on different ER random graphs with edge probabilities of 0.5. **d** Assessing the effect of different α values on the overall performance on different ER random graphs with edge probabilities of 0.7. The error bars display the variance obtained through sampling over three non-isomorphic instances

with varying α values. As illustrated in Fig. 8, on a random graph (ER graph) with six nodes and a degree of 3, we sampled the final states of the quantum circuits corresponding to the these α value combinations. Figure 8 shows the measurement count statistics and the final total expectation value. Among them, we indicate the α values for each different configuration, and our CVaR aims to use Eq. (7) to replace the original objective function of the circuit for the next iteration optimization, in order to obtain the overall expectation value $F(z)$ of the final circuit. We observed that as the α value decreases, the overall measurement state counts of the circuit tend to concentrate on states with lower expectation values. Concurrently, the overall expectation values of CVaR optimization for the three distinct α configurations are -5.886, -6.513 and -7.0, respectively. This further suggests that at lower α values, the overall expectation values of the quantum circuits are more heavily concentrated toward lower eigenvalues, thereby exhibiting superior performance. These observations not only validate the effectiveness of our method but also provide strong evidence from the overall distribution of circuit measurement states.

By adjusting the α parameter, we are able to dynamically control the expressiveness of the ansatz in the quantum circuit. The experimental results clearly indicate that finely

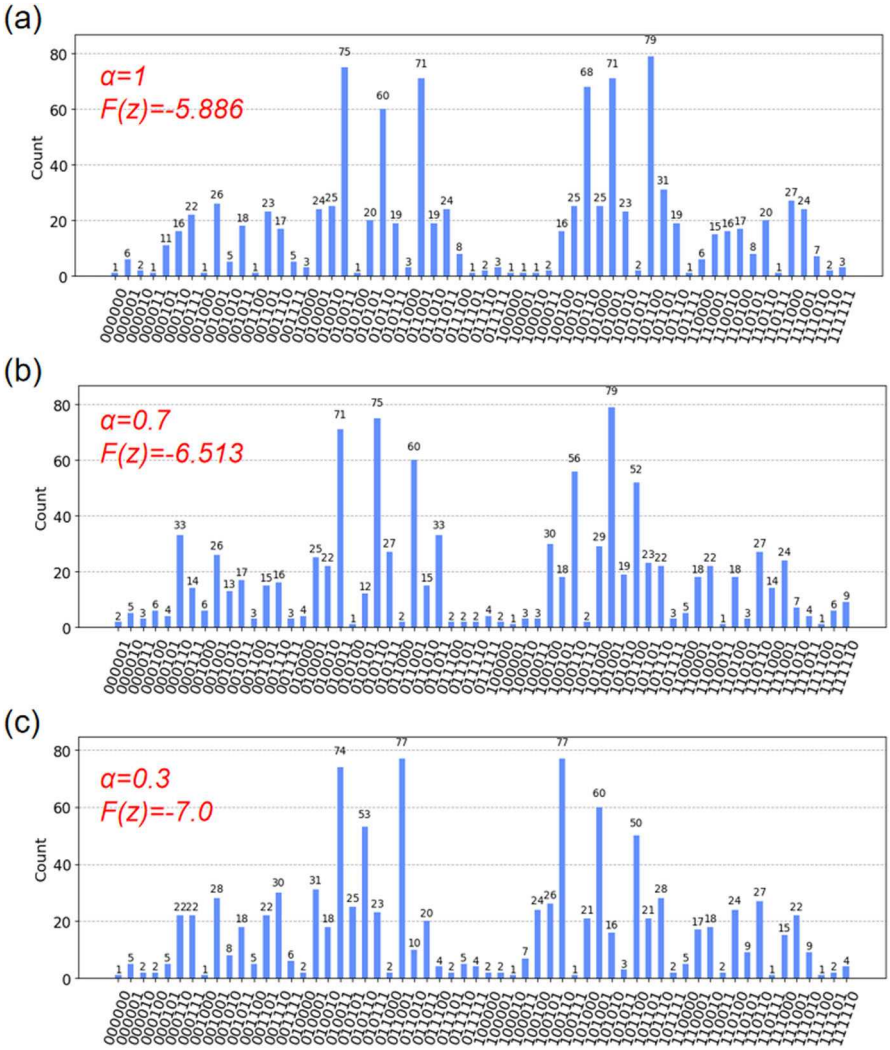

Fig. 8 Count the number of times the final quantum circuit measures each state corresponding to the α values of the three configurations, $F(z)$ represents the overall expectation value after CVaR optimization

tuning the α parameter can significantly enhance the performance of the quantum circuit in addressing specific optimization challenges. This study further highlights the critical role of the parameter α in the CVaR optimization process, substantially improving the efficiency and precision of quantum algorithms in practical applications.

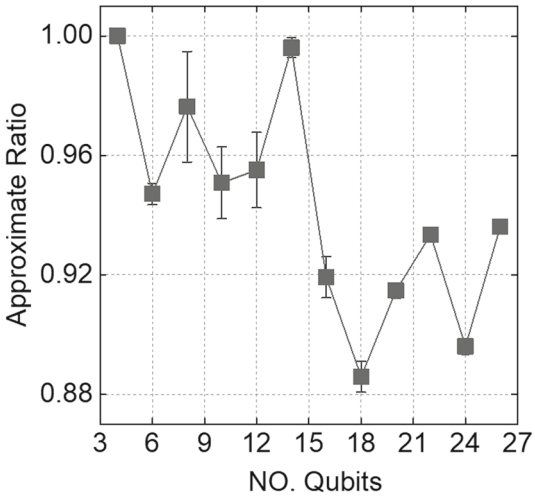

Fig. 9 Quantum approximation ratio obtained using classical simulation methods of the CNN-CVaR-QAOA algorithm under different qubit configurations. Throughout this process, we maintained a fixed circuit depth of 3 and ensured that the graphs involved had a degree of 3. With this setup, we were able to systematically investigate the algorithm's performance across various quantum bit scales

4.3 Applicability of our method

In this section, we conducted an in-depth investigation of the application of the method in larger-scale quantum systems, particularly addressing scalability and computational complexity issues.

As shown in Fig. 9, we fixed the circuit depth at 3 and tested systems with 4–22 quantum bits on a 3-regular graph. The experimental results show that the error bars reflect the variance of the approximation ratio obtained by running each instance independently three times. It is worth noting that even on a 26-qubit system, our method still maintained a high approximation ratio of 93%, indicating that as the number of qubits increases, the performance of our proposed method remains robust.

These data clearly demonstrate that our method not only performs well on small-scale problems but also maintains its effectiveness as the problem size increases. This is of great significance for solving NP-hard problems in the real world. However, it is important to note that the increase in the number of qubits also leads to an increase in computational overhead. Specifically, as the number of quantum bits increases, adding more qubits will significantly impact the time complexity of this method, meaning that the computational time required to optimize the entire quantum circuit system will increase significantly. This phenomenon reveals that there is a trade-off between scalability and computational efficiency in the field of quantum computing, which is a key issue that needs to be further addressed in our future research.

Then, we have extended our evaluation to include another combinatorial optimization problem: the exact cover problem. The exact cover problem, given a collection of subsets S from a target set X, seeks to identify a sub-collection S^* such that each element in X appears exactly once in one of the subsets within S^*. In other words, the goal is to find a set of pairwise disjoint subsets whose union equals X [47].

As illustrated in Fig. 10, our collection consists of elements X_1 and X_2. We employ

A

Target Set		Subset Collection
X_1 = { 1, 2, 3, 4, 5, ..., 23}	S_1:	[{1,3,5,6,8,10,13,14,16,18,20,22}, {0,2,4,7,9,11,12,15,17,19,21,23}, {0,2,4,7,8,11,13,15,16,18,20,22}, {1,3,5,6,9,10,12,14,17,19,21,23}, {0,2,4,6,8,10,12,14,17,19,21,23}, {1,3,5,7,9,11,13,15,16,18,20,22}]
Solutions: 3		

B

Target Set		Subset Collection
X_2 = { 1, 2, 3, 4, 5, ..., 23}	S_2:	[{1,3,5,6,8,11,13,15,16,19,21,23}, {0,2,4,7,9,10,12,14,17,18,20,22}, {0,2,4,7,8,11,13,15,16,18,20,23}, {1,3,5,6,9,10,12,14,17,19,21,22}, {0,2,4,7,9,11,12,15,17,19,21,23}, {1,3,5,6,8,10,13,14,16,18,20,22}, {0,2,4,7,9,11,12,14,16,18,20,22}, {1,3,5,6,8,10,13,15,17,19,21,23}, {1,3,5,7,9,11,13,15,17,19,20,22}, {0,2,4,6,8,10,12,14,16,18,21,23}, {0,2,4,7,9,11,13,15,17,19,21,23}, {0,2,4,6,8,10,12,14,17,18,20,22}, {1,3,5,6,9,11,13,14,16,18,20,22}, {0,2,4,7,9,11,13,15,16,18,20,23}]
Solutions: 5		

Fig. 10 Two examples of precise coverage instances (X_1 S_1 and $X_2 S_2$) are provided. Request to find a solution for $X_1 S_1$ containing subsets and a solution for $X_2 S_2$ containing subsets

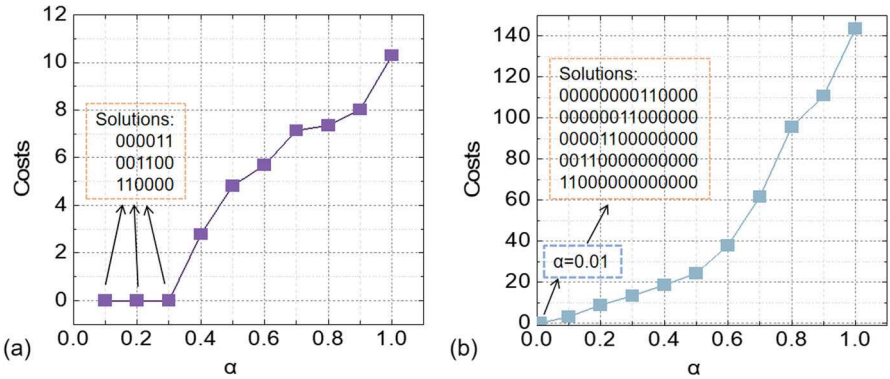

Fig. 11 **a** In set S1, run CNN-CVaR and observe the changes in the cost function corresponding to different α. **b** In set S2, run CNN-CVaR and observe the changes in the cost function corresponding to different α

the CNN-CVaR strategy to identify the corresponding sub-collection solutions. Subsequently, we apply this strategy to tackle this type of combinatorial optimization problem, with the results presented in Fig. 11. It is noteworthy that when α=1, the cost function does not reach its minimum. By reducing the value of α, we can enhance system performance and ultimately arrive at the optimal solution.

In Problem A, S_1 comprises 6 subsets, corresponding to a quantum approximate optimization circuit with 6-qubit. Our objective is to identify three nonintersecting subsets from S_1 that are all elements of X_1. Figure 11a demonstrates that when α is reduced to 0.3, the system successfully converges to the optimal cost value, thereby finding the best solution. In Problem B, we increase the number of subsets to 14 and aim to find five solutions. In this scenario, the system also fails to converge to the minimum value when $\alpha = 0.1$. However, when we set α to 0.01, we observe that the system is capable of finding the top five solutions. We dynamically control the expressiveness of ansatz by adjusting the value of α.

These findings demonstrate that our CNN-CVaR predictive model is capable of learning and utilizing the inherent correlations within different combinatorial optimization problems. By dynamically adjusting the expressiveness of the ansatz through the parameter α, we can fine-tune the performance of system and obtain optimal solutions in a wide range of problem settings.

5 Conclusions

In this study, we introduce CNN-CVaR-QAOA, a novel approach that integrates classical CNN architecture with CVaR aggregation to tackle the Max-Cut problem on small-scale quantum devices. Our method predicts QAOA parameters for each depth using a CNN network structure and incorporates the CVaR strategy as the cost function for the quantum circuit. Through comprehensive benchmark tests on diverse problem sets, we have demonstrated the superior performance of the CNN-CVaR-QAOA strategy compared to random initialization methods and interpolation techniques. Our numerical results reveal compelling advantages, particularly evident in graph instances, where the depth-2 circuit of our approach exhibits performance levels akin to pure CNN optimization at a depth of 6. Notably, we observe that the α parameter in the CNN-CVaR-QAOA strategy plays a significant role in performance enhancement, with smaller α values emphasizing favorable observation samples, thereby leading to a smoother objective function and improved approximation ratios.

One of the distinguishing features of our proposed strategy is its ability to personalize predicted parameters for individual graph instances, enabling a broader exploration of optimal parameters. This flexibility becomes increasingly valuable as quantum computing hardware progresses toward handling larger-scale QAOA instances requiring $p>>1$. Our method's potential to match the approximation ratio of original QAOA at lower depths positions it as a compelling alternative in the current NISQ era, characterized by the emergence of larger quantum devices grappling with high noise levels for executing deep circuits. While our focus lies on addressing the Max-Cut problem, we envision broader applicability of our method across various optimization problems. As quantum computers evolve and scale up in qubit count, the practicality of our approach becomes more pronounced, particularly given the finite constraints on the number of layers in QAOA for achieving feasible solutions. The CNN-CVaR-QAOA strategy represents a promising advancement in quantum optimization techniques, offering enhanced performance and adaptability that hold relevance not only for the Max-Cut problem but for a wide array of optimization challenges in the quantum computing landscape.

Author Contributions All authors contributed to the study conception and design.

Funding This work is supported by the National Natural Science Foundation of China (Grant Nos. 61874001, 62004001, 62201005, 62004001, 62304001), the Anhui Provincial Natural Science Foundation under Grant Nos. 2308085QF213, 2308085QF195, and the Natural Science Research Project of Anhui Educational Committee under Grant Nos. 2022AH050106, 2023AH050072.

Data Availability The data that support the findings of this study are available on request from the corresponding author upon reasonable request.

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